

G.E.T.S. Cash vs. Future

Arbitrage Quote Model

User Manual

Version 1.0.0.1

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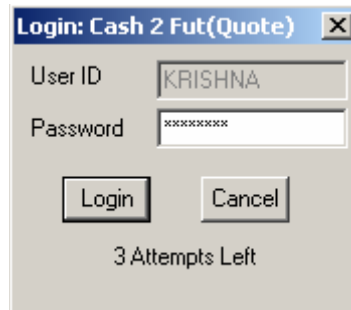
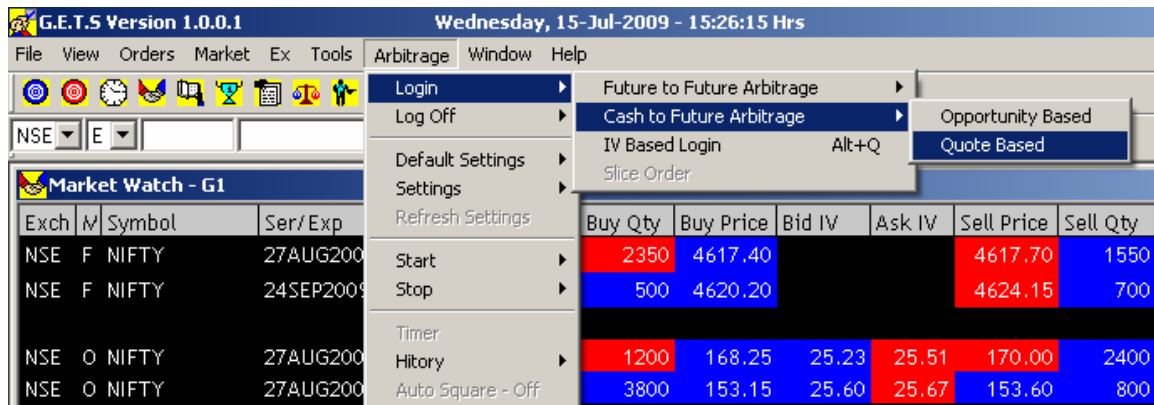
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Cash vs. Future arbitrage quote model is used for gaining the advantage of volatile pricings between cash and future markets. The advantage of quote model is to reach the more accurate spread expected and faster execution in capturing the opportunity.

I. Login:

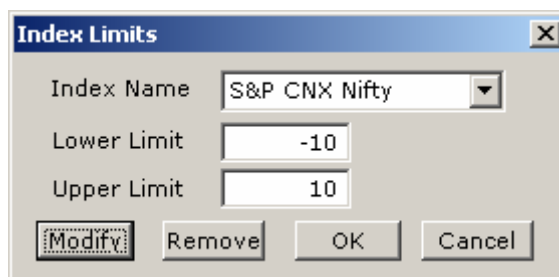
The user needs to login through GETS CTCL client, after which he will get the access of Cash vs. future quote model from Arbitrage menu as shown below.



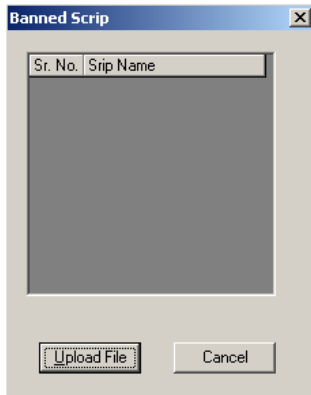
If the user enters wrong password for more than 3 times, his login will be blocked. To unblock, he has to request the system administrator.

During the login sequence to Cash vs. future arbitrage quote model, following checks must be defined, which will be validated in the post-login and execution.

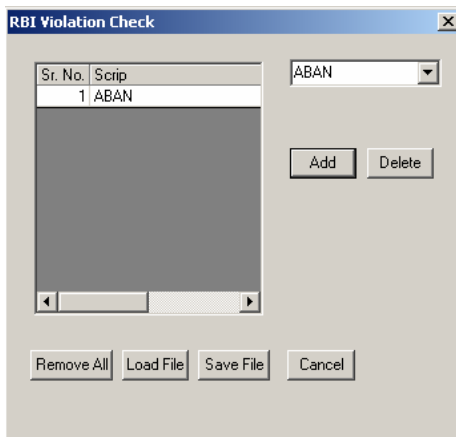
Index Limits Check – When index crosses the set volatility range, then automated arbitrage will be stopped. Need to define the index and volatility range.



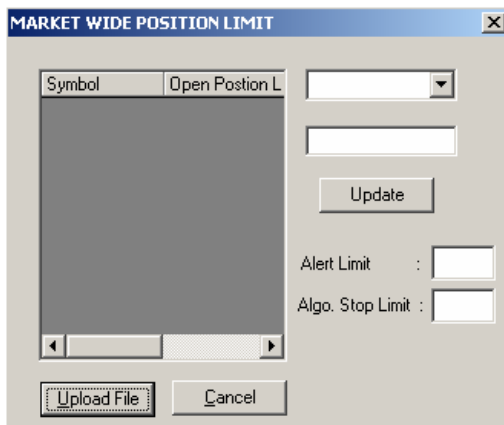
Banned Scrip Check – Trading in any banned scrip can be ruled out by uploading the banned scrip list. All violated symbols won't be loaded in the arbitrage settings form.



RBI Violation Check – Trading in any RBI violated scrip can be ruled out by uploading RBI violation list. All violated symbols won't be loaded in the arbitrage settings form.



Market Wide Position Limit Check – Trading in contracts, where market wide position limit reached, will be ruled out by uploading the market wide position limit list. All violated symbols won't be loaded in the arbitrage settings form.



II. Input parameters:

After successful login, the user will be prompted with the cash to future limit settings dialog as shown below. He can call the dialog from market watch using short key SHIFT+F7.

All input parameters are explained below.

1. Exchange: NSE
2. Symbol: Derivative symbols listed on NSE
3. PRO/CLI: Choose for proprietary trading / client trading
4. Account ID: Select the client account ID
5. Participant ID: NSE participant ID, in case of institutional client
6. Expiry: Derivative expiry date
7. Lot Size: Lot size of the symbol will be selected automatically

Settings for Buy Cash & Sell Future (Fresh Arbitrage): Check Buy Cash :: Sell Future

8. Difference: Input required absolute spread for Fresh Arbitrage
9. Order Lots: Input the no. of lots that has to be placed for each order
10. Total Lots: Input the total no. of lots to execute
11. Trd Qty: Displays the traded quantity for selected symbol
12. Remaining Qty: Displays the remaining quantity for selected symbol

Settings for Sell Cash & Buy Future (Unwind): Check Sell Cash :: Buy Future

- 13. Difference: Input required absolute spread for Unwind
- 14. Order Lots: Input the no. of lots that has to be placed for each order
- 15. Total Lots: Input the total no. of lots to execute
- 16. Trd Qty: Displays the traded quantity for selected symbol
- 17. Remaining Qty: Displays the remaining quantity for selected symbol

First Order Base Settings

- 18. First Order: Options – Cash Based / Buy / Sell – Select any one of these

Cash Based – On selecting this option, cash order will be placed first and after full execution of the cash order, future order will be placed.

Fresh Arbitrage: Buy cash order will be submitted and after the cash quantity is bought, sell future order will be placed to exchange.

Unwind: Sell cash order will be submitted and after the cash quantity is sold, buy future order will be placed to exchange.

Buy – On selecting this option, after buy order gets fully traded, sell order will be sent to exchange.

Sell – On selecting this option, after sell order gets fully traded, buy order will be sent to exchange.

- 19. First Order: Options – All Buy / All Sell / All Cash / Default – Select any one of these

In this condition, if All Buy / All Sell / All Cash option is selected, then this option will precede the condition selected in 18th point, irrespective of Fresh Arbitrage / Unwind. If selected default, the execution will be done as per settings in 18th point. This condition applies across symbols being input for Fresh Arbitrage & Unwind.

All Buy – On selecting this option, after buy order gets fully traded, sell order will be sent to exchange.

All Sell – On selecting this option, after sell order gets fully traded, buy order will be sent to exchange.

All Cash – On selecting this option, cash order will be placed first and after full execution of the cash order, future order will be placed.

Default – On selecting this option, execution will be done as per selection in 18th point.

Pending Order related settings

- 20. Best Ask/Bid – Input the no. of modifications allowed. If the against order is pending, then it will be modified for the no. of times mentioned in this text box.

21. Fishing Val. – Input the compromise rate, which will be added / subtracted from the pending order price based on the transaction type (Buy/Sell). Value should be in the multiples of 0.05.

22. Time – Input the time interval to modify the pending against order.

23. Mkt Ord Upto – Input the maximum price tick by which the pending order price will be modified. If this is defined as zero, pending order will be modified as market order.

24. Depth Upto – Input the market picture depth size consideration for weighted average price calculation. Value should be in the range of 1 to 5.

Other settings

25. Th Qty % – Input the threshold quantity to be considered for cash order generation, which is required in validating the quantity (percentage to lot size) availability in the market depth.

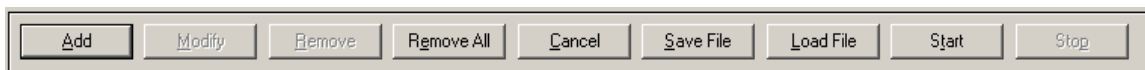
26. Disc. Qty % - Input the disclosed quantity percentage, which is considered for counter cash orders. If the against order is cash order, then disclosed quantity is not considered.

27. Abs Qty % - Input the absolute quantity percentage. If the absolute quantity percentage of the pending quantity of counter order gets traded, then against order will be submitted. If the counter order is of future, then this check is not considered.

28. Second Order Actual diff – If this option is checked, then the price of against order will be calculated considering the avg. trade price of counter order and difference set. If this is not checked, the price of against order will be calculated considering the order price of counter order which is fully traded and difference set.

29. Opportunity base – If this option is checked, then both the arbitrage orders (cash & future) will be placed; whenever there is on screen opportunity is observed.

III. Controlling buttons:



Add: To add the new arbitrage opportunity setting to the arbitrage farm

Modify: To modify any previously added settings to the arbitrage farm

Remove: To remove any previously added settings from the arbitrage farm

Remove All: Removes all arbitrage settings from the arbitrage farm

Cancel: To cancel any partially filled arbitrage setting

Save File: To save all the arbitrage settings present in the arbitrage farm

Load File: To load previously saved arbitrage settings to arbitrage farm

Start: To start calculating the opportunity and submission of the orders

Stop: To stop calculating the opportunity. Calling this will cancel all pending orders from order book operated by the arbitrage module.

At any point of time the user can add maximum of 20 symbols and trading happens simultaneously across all counters.

Reviewed by: Hitesh Hakani

In case of multiple lots, if the pending order quantity not available in the market depth on futures bid side, then the cash order will be canceled and fresh counter cash order will be placed based on the total future bid quantity & market depth upto.

Modified cash order price - Current Future Best Bid minus difference set

Counter cash order is partially traded:

Conditions checked: Absolute quantity & Timer

If traded quantity is below absolute quantity % till the set time interval – Cancel the pending buy cash order and traded cash quantity will be reversed at market price.

In case of multiple order lots, pending buy order quantity will be modified to the nearest lot size on higher side and order price to market. And places the future sell order with cash side traded quantity as order quantity.

If executed quantity is above absolute quantity % - Future order will be generated

Pending counter buy cash order after against sell future order is placed:

Modify the pending cash order to market after set time interval is finished.

Counter cash order is fully traded:

Future order will be generated considering the below parameters.

Order Quantity – traded quantity counter buy cash order

Order Price (if Second order actual difference checked) – Avg. trade price of cash order executed + Set difference

Order Price (if Second order actual difference not checked) – Order price of cash order executed + Set difference

Against Sell Future Order Cases

Against future order is pending / partially traded:

Conditions checked: Best Bid/Ask modifications, Market Order Upto, Time interval

Pending future sell order will be modified to best bid price for the no. of best bid/ask modifications mentioned after the set time interval. If the order is pending even after the no. of modifications, then the pending future order will be modified as below.

If Market Order Upto value is zero - modify the pending future order as market order.

If Market Order Upto value is other than zero - modify the pending future sell order price by subtracting the set market order upto value from pending order price. If this order doesn't get fully traded, same needs to be closed with manual intervention.

Against future order is fully traded:

Order cycle completed. If remaining quantity more than zero, Case I will be repeated.

Case II: Buy Cash :: Sell Future with first order base option as Sell

After filling the input parameters, when the user call the Start control button first sell future counter order will be placed on satisfying the below conditions (for single / multiple order lots).

Remaining quantity – should be more than zero

Depth upto & Threshold quantity – Total cash ask quantity, considering the set market depth upto level, should be more than or equal to set % of threshold quantity

And the counter future order parameters will be as follows.

Order Quantity – Set order lots considering the remaining quantity

Order Price – Current Cash Best Ask plus difference set

Counter sell future order cases

Counter future order is pending

Conditions checked: Difference set, Market Depth Upto

For every tick change in the cash best ask price, the price of pending future sell order will be modified, considering the set difference.

In case of multiple lots, if the pending order quantity is not available in the market depth on cash ask side, then the future order will be canceled and fresh counter future order will be placed based on the total cash ask quantity & market depth upto.

Modified future order price - Current Cash Best Ask plus difference set

Counter future order is partially traded (in case of multiple lots)

Cancel the pending future sell order and place against cash buy order.

Counter future order is fully traded

Against cash buy order will be generated considering the below parameters.

Order Quantity – traded quantity of counter sell future order
Order Price (if Second order actual difference checked) – Avg. trade price of future sell order executed – Set difference
Order Price (if Second order actual difference not checked) – Order price of future sell order executed – Set difference

Against buy cash order cases

Against cash order is pending / partially traded

Conditions checked: Best Bid/Ask modifications, Market Order Upto, Time interval

Pending cash buy order will be modified to best ask price for the no. of best bid/ask modifications mentioned after the set time interval. If the order is pending even after the no. of modifications, then the pending cash order will be modified as below.

If Market Order Upto value is zero - modify the pending cash order as market order.
If Market Order Upto value is other than zero - modify the pending cash buy order price by adding the set market order upto value from pending order price. If this order doesn't get fully traded, same needs to be closed with manual intervention.

Against cash order is fully traded

Order cycle completed. If remaining quantity more than zero, Case II will be repeated.

Case III: Sell Cash :: Buy Future with first order base option as Sell / Cash Based

After filling the input parameters, when the user call the Start control button, first sell cash counter order will be placed on satisfying the below conditions.

In case of single order lot,

Remaining quantity – should be more than zero

In case of multiple order lots,

Remaining quantity – should be more than zero and
Depth upto & Threshold quantity – Total future ask quantity, considering the set market depth upto level, should be more than or equal to set % of threshold quantity

And the order parameters will be as follows.

Order Quantity – Set order lots considering the remaining quantity
Order Price – Current Future Best Ask plus difference set
Disclosed quantity – Set % of disclosed quantity

Counter sell cash order cases

Counter cash order is pending:

Conditions checked: Difference set, Market Depth Upto

For every tick change in the future best ask price, the price of pending cash sell order will be modified, considering the set difference.

In case of multiple lots, if the pending order quantity not available in the market depth on futures ask side, then the cash order will be canceled and fresh counter cash order will be placed based on the total future ask quantity & market depth upto.

Modified cash order price - Current Future Best Ask plus difference set

Counter cash order is partially traded:

Conditions checked: Absolute quantity & Timer

If traded quantity is below absolute quantity % till the set time interval – Cancel the pending sell cash order and traded cash quantity will be reversed at market price.

In case of multiple order lots, pending sell order quantity will be modified to the nearest lot size on higher side and order price to market. And places the future buy order with cash side traded quantity as order quantity.

If executed quantity is above absolute quantity % - Future order will be generated

Pending counter sell cash order after against buy future order is placed:

Modify the pending cash order to market after set time interval is finished.

Counter cash order is fully traded:

Future order will be generated considering the below parameters.

Order Quantity – traded quantity counter sell cash order

Order Price (if Second order actual difference checked) – Avg. trade price of cash order executed – Set difference

Order Price (if Second order actual difference not checked) – Order price of cash order executed – Set difference

Against Buy Future Order Cases

Against future order is pending / partially traded:

Conditions checked: Best Bid/Ask modifications, Market Order Upto, Time interval

Pending future buy order will be modified to best ask price for the no. of best bid/ask modifications mentioned after the set time interval. If the order is pending even after the no. of modifications, then the pending future order will be modified as below.

If Market Order Upto value is zero - modify the pending future order as market order.

If Market Order Upto value is other than zero - modify the pending future buy order price by adding the set market order upto value from pending order price. If this order doesn't get fully traded, same needs to be closed with manual intervention.

Against future order is fully traded:

Order cycle completed. If remaining quantity more than zero, Case III will be repeated.

Case IV: Sell Cash :: Buy Future with first order base option as Buy

After filling the input parameters, when the user call the Start control button first buy future counter order will be placed on satisfying the below conditions (for single / multiple order lots).

Remaining quantity – should be more than zero

Depth upto & Threshold quantity – Total cash bid quantity, considering the set market depth upto level, should be more than or equal to set % of threshold quantity

And the counter future order parameters will be as follows.

Order Quantity – Set order lots considering the remaining quantity

Order Price – Current Cash Best Bid minus difference set

Counter buy future order cases

Counter future order is pending

Conditions checked: Difference set, Market Depth Upto

For every tick change in the cash best bid price, the price of pending future buy order will be modified, considering the set difference.

In case of multiple lots, if the pending order quantity is not available in the market depth on cash bid side, then the future order will be canceled and fresh counter future order will be placed based on the total cash bid quantity & market depth upto.

Modified future order price - Current Cash Best Bid minus difference set

Counter future order is partially traded (in case of multiple lots)

Cancel the pending future buy order and place against cash sell order.

Counter future order is fully traded

Against cash sell order will be generated considering the below parameters.

Order Quantity – traded quantity of counter buy future order

Order Price (if Second order actual difference checked) – Avg. trade price of future buy order executed + Set difference

Order Price (if Second order actual difference not checked) – Order price of future buy order executed + Set difference

Against sell cash order cases

Against cash order is pending / partially traded

Conditions checked: Best Bid/Ask modifications, Market Order Upto, Time interval

Pending cash sell order will be modified to best bid price for the no. of best bid/ask modifications mentioned after the set time interval. If the order is pending even after the no. of modifications, then the pending cash order will be modified as below.

If Market Order Upto value is zero - modify the pending cash order as market order.

If Market Order Upto value is other than zero - modify the pending cash sell order price by subtracting the set market order upto value from pending order price. If this order doesn't get fully traded, same needs to be closed with manual intervention.

Against cash order is fully traded

Order cycle completed. If remaining quantity more than zero, Case IV will be repeated.

V. Order Book & Trade Book:

All the orders resulting from the execution of cash vs. future arbitrage quote model gets updated in pending order book (F3). And all trades get updated in previous trade book (F8).

Pending Orders Of All Scripts																				
GreekC	RefNo.	Exch	Trader	Inst	Na	Code	Symb	Ser/Exp/Grc	Strik	Op	B/S	Pend	Price	Total	Prc	Clie	Client N.	Exch. Order No.	Settlor	Valid
AJAY1	200002808	NSE	18192	F	FUTIDX	8986	NIFTY	27AUG2009		Buy	50	0.00	50	CLI 0H01	HITESH	2009080310000122	07782	DAY		
AJAY1	200002808	NSE	18192	F	FUTIDX	5999	NIFTY	24SEP2009		Sell	50	0.00	50	CLI 0H01	HITESH	2009080310000122	07782	DAY		
AJAY1	200002807	NSE	18192	F	FUTIDX	8986	NIFTY	27AUG2009		Buy	50	0.00	50	CLI 0H01	HITESH	2009080310000121	07782	DAY		
AJAY1	200002807	NSE	18192	F	FUTIDX	5999	NIFTY	24SEP2009		Sell	50	0.00	50	CLI 0H01	HITESH	2009080310000121	07782	DAY		
AJAY1	200002806	NSE	18192	F	FUTIDX	8986	NIFTY	27AUG2009		Buy	50	0.00	50	CLI 0H01	HITESH	2009080310000120	07782	DAY		
AJAY1	200002806	NSE	18192	F	FUTIDX	5999	NIFTY	24SEP2009		Sell	50	0.00	50	CLI 0H01	HITESH	2009080310000120	07782	DAY		
AJAY1	200002811	NSE	18192	F	FUTIDX	8986	NIFTY	27AUG2009		Buy	50	0.00	50	CLI 0H01	HITESH	2009080310000125	07782	DAY		
AJAY1	200002811	NSE	18192	F	FUTIDX	5999	NIFTY	24SEP2009		Sell	50	0.00	50	CLI 0H01	HITESH	2009080310000125	07782	DAY		
AJAY1	200002810	NSE	18192	F	FUTIDX	8986	NIFTY	27AUG2009		Buy	50	0.00	50	CLI 0H01	HITESH	2009080310000124	07782	DAY		
AJAY1	200002810	NSE	18192	F	FUTIDX	5999	NIFTY	24SEP2009		Sell	50	0.00	50	CLI 0H01	HITESH	2009080310000124	07782	DAY		
												BuyQty : 3100		SellQty : 0						

Trades-All																			
Exch	Membr	TraderC	Inst	Na	Symbol/Sci	Ser/Exp/Grc	Strike	Opt	B/S	Qty	Price	Pro	Exch. Order No	Trader	Time	GreekC	Ref No		
NSE	07782	18192	F	FUTIDX	NIFTY	24SEP2009		Sell	50	4691.70	CLI	09080410000044	20019	12:13:50	AJAY1	100000071			
NSE	07782	18192	F	FUTIDX	NIFTY	24SEP2009		Sell	50	4691.70	CLI	09080410000044	20018	12:13:50	AJAY1	100000071			
NSE	07782	18192	F	FUTIDX	NIFTY	24SEP2009		Sell	50	4691.70	CLI	09080410000044	20017	12:13:50	AJAY1	100000071			
NSE	07782	18192	F	FUTIDX	NIFTY	27AUG2009		Buy	50	4689.00	CLI	09080410000040	20016	12:13:44	AJAY1	100000067			
NSE	07782	18192	F	FUTIDX	NIFTY	24SEP2009		Sell	50	4692.35	CLI	09080410000042	20015	12:13:42	AJAY1	100000069			
NSE	07782	18192	F	FUTIDX	NIFTY	27AUG2009		Buy	50	4689.25	CLI	09080410000041	20014	12:13:42	AJAY1	100000068			
NSE	07782	18192	O	OPTIDX	NIFTY	27AUG2009	4600.00	PE	Sell	50	128.30	CLI	09080410000039	20013	12:13:18	AJAY1	100000066		
NSE	07782	18192	O	OPTIDX	NIFTY	27AUG2009	4600.00	CE	Buy	50	214.90	CLI	09080410000038	20012	12:13:16	AJAY1	100000065		
NSE	07782	18192	F	FUTSTK	RELIANCE	27AUG2009		Buy	150	2043.70	CLI	09080410000037	20010	12:10:55	AJAY1	100000064			
NSE	07782	18192	F	FUTSTK	RELIANCE	27AUG2009		Sell	150	2045.85	CLI	09080410000032	20004	12:10:14	AJAY1	100000059			
NSE	07782	18192	F	FUTSTK	RELIANCE	27AUG2009		Buy	150	2046.00	CLI	09080410000031	20003	12:10:04	AJAY1	100000058			

VI. Net Positions:

All the position summary of previous trades updated in net position dialog (ALT+F6)

Net Position																			
Exchange	Mkt Type	Symbol	Periodicity	User	Security / Contract														
NSE	All	All	Daily	AJAY1															
Exch	User	Client	MI	Instrument	Symbol	Ser/Exp	Strike	P	Op	Buy Qt	Buy Val	Buy Avg	Sell Qt	Sell Val	Sell Avg	Net	Net Val	MTM	G/L
NSE	AJAY1	07782	O	OPTIDX	NIFTY	27AUG2009	4600.00	CE		50	10745.00	214.90	0	0.00	0.00	50	10745.00	357.50	
NSE	AJAY1	07782	O	OPTIDX	NIFTY	27AUG2009	4600.00	PE		0	0.00	0.00	50	6415.00	128.30	-50	-6415.00	237.50	
NSE	AJAY1	07782	F	FUTIDX	NIFTY	27AUG2009				100	468912.50	4689.13	0	0.00	0.00	100	468912.50	1257.00	
NSE	AJAY1	07782	F	FUTSTK	NIFTY	24SEP2009				0	0.00	0.00	550	2580467.50	4691.76	-550	2580467.50	-7282.00	
NSE	AJAY1	07782	F	FUTSTK	RELIANCE	27AUG2009				300	613455.00	2044.85	150	306877.50	2045.85	150	306577.50	532.50	
Tota										450	093112.50		750	2893760.00		-300	1800647.50	-4897.50	

G.E.T.S.	Net Position	NIFTY09AUG4600CE	Buy Qty: 50	Sell Qty: 0	Net Qty: 50	Buy Val: 10745.00	Sell Val: 0.00	Net Val: 10745.00	MTM: 405.00
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